

EE 133 Final Exam – Complete Solutions

Winter 2026

Constants: $n_i = 10^{10} \text{ cm}^{-3}$, $kT = 0.026 \text{ eV}$, $q = 1.602 \times 10^{-19} \text{ C}$, $\varepsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$, $\varepsilon_{Si} = 11.8\varepsilon_0$, $\varepsilon_{ox} = 3.9\varepsilon_0$, $E_{G,Si} = 1.12 \text{ eV}$

Problem 1

$N_D = 10^{17} \text{ cm}^{-3} \gg n_i$, so charge neutrality gives $n \approx N_D$:

$$n = 10^{17} \text{ cm}^{-3}, \quad p = \frac{n_i^2}{n} = \frac{10^{20}}{10^{17}} = 10^3 \text{ cm}^{-3}$$

Problem 2

(a) Fermi Level

In equilibrium E_F is constant and flat everywhere – horizontal line through the marked dot at $z = 0$.

(b) Electric Field Direction

$$\mathcal{E} = \frac{1}{q} \frac{dE_i}{dz} > 0 \quad \Rightarrow \quad \mathcal{E} \text{ points in the } +z \text{ direction.}$$

(c) Electron Diffusion Current

$n = n_i e^{(E_F - E_i)/kT}$. E_F flat, E_i rises toward $+z \Rightarrow n$ decreases toward $+z$:

$$J_{N\text{-diff}} = qD_n \frac{dn}{dz} < 0 \quad \Rightarrow \quad J_{N\text{-diff}} \text{ in the } (-z) \text{ direction.}$$

(d) Hole Diffusion Current

$p = n_i e^{(E_i - E_F)/kT}$ increases toward $+z$:

$$J_{H\text{-diff}} = -qD_p \frac{dp}{dz} < 0 \quad \Rightarrow \quad J_{H\text{-diff}} \text{ in the } (-z) \text{ direction.}$$

(e) Hole Drift Current

$$J_{H\text{-drift}} = qp\mu_p\mathcal{E} > 0 \quad \Rightarrow \quad J_{H\text{-drift}} \text{ in the } (+z) \text{ direction.}$$

(f) Electron Drift Current

$$J_{N\text{-drift}} = qn\mu_n\mathcal{E} > 0 \quad \Rightarrow \quad J_{N\text{-drift}} \text{ in the } (+z) \text{ direction.}$$

(g) Total Current

Zero. This is an equilibrium system; drift and diffusion cancel exactly for each carrier species, which is the condition that enforces a flat E_F .

(h) Electron Density at $z = 0$

The dot lies on E_i , so $E_F = E_i$ at $z = 0$:

$$n = n_i e^{(E_F - E_i)/kT} = n_i = 10^{10} \text{ cm}^{-3}$$

(i) Hole Density at $z = 0$

$$p = n_i e^{(E_i - E_F)/kT} = n_i = 10^{10} \text{ cm}^{-3}$$

Problem 3

Using $J_n = n\mu_n \frac{dF_n}{dx}$ and $J_p = p\mu_p \frac{dF_p}{dx}$:

(a) Electron Current?

F_n is flat $\Rightarrow dF_n/dx = 0 \Rightarrow J_n = 0$. **No electron current.**

(b) Hole Current?

F_p has a positive slope $\Rightarrow dF_p/dx > 0 \Rightarrow J_p = p\mu_p(dF_p/dx) > 0$.

Yes. Hole current in the $+x$ direction.

Problem 4

For an n^+p diode, injection into the p-side dominates:

$$J_0 \approx \frac{qD_n n_i^2}{L_n N_A} \propto \frac{1}{N_A}$$

J_0 is independent of N_D (n-side already n^+ , minority carrier density negligible).

- (a) Double N_D : no significant effect.
- (b) Halve N_D : no significant effect.
- (c) Double N_A : $J_0 \rightarrow J_0/2$. Halves current.
- (d) Halve N_A : $J_0 \rightarrow 2J_0$. **Doubles current.** ✓

Answer: (d)

Problem 5

$$C_j(V_A) = C_{j0} \left(1 - \frac{V_A}{V_{bi}} \right)^{-1/2}$$

$$1 = 2 \left(1 - \frac{V_A}{0.7} \right)^{-1/2} \Rightarrow \left(1 - \frac{V_A}{0.7} \right)^{1/2} = 2 \Rightarrow 1 - \frac{V_A}{0.7} = 4 \Rightarrow V_A = -2.1 \text{ V}$$

$V_A = -2.1 \text{ V}$

Problem 6

Heavily implant the Si directly beneath the metal to $N_D \gtrsim 10^{19}\text{--}10^{20} \text{ cm}^{-3}$ (create an n^+ contact region). The depletion width $W \propto 1/\sqrt{N_D}$ becomes thin enough ($\sim$ few nm) for carriers to quantum-mechanically tunnel through the Schottky barrier (field emission), producing ohmic behavior.

Problem 7

(a) Region A – Reverse Bias, $n = 2$

Generation-Recombination (G-R) current via SRH traps in the depletion region: $I \propto n_i \exp(qV/2kT)$.

(b) Region B – Forward Bias, $n = 1$

Ideal diffusion (Shockley) current: minority carriers injected and recombine in the quasi-neutral regions, $I = I_0 \exp(qV/kT)$.

(c) Region C – Forward Bias, $n = 2$

SRH recombination within the depletion region dominates at higher forward bias: $I \propto n_i \exp(qV/2kT)$.

(d) Region D – Flattening

Series resistance. At high current, the voltage drop IR_s across the bulk resistance limits further exponential growth.

(e) Si or Ge?

Silicon. Turn-on voltage $\approx 0.6\text{--}0.7$ V is consistent with Si ($E_G = 1.12$ eV). Ge turns on at $\approx 0.2\text{--}0.3$ V.

Problem 8

$T_1 = 233$ K, $T_2 = 398$ K. With D , L , E_G fixed, $I_0 \propto n_i^2 \propto T^3 \exp(-E_G/kT)$, so:

$$I(T) \propto T^3 \exp\left(\frac{qV_A - E_G}{kT}\right)$$

$$\frac{I(T_2)}{I(T_1)} = \left(\frac{T_2}{T_1}\right)^3 \exp\left[\frac{qV_A - E_G}{k} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right]$$

$$\left(\frac{398}{233}\right)^3 = (1.708)^3 = 4.99$$

$$qV_A - E_G = 0.7 - 1.12 = -0.42 \text{ eV}$$

$$\frac{1}{T_2} - \frac{1}{T_1} = \frac{1}{398} - \frac{1}{233} = -1.779 \times 10^{-3} \text{ K}^{-1}$$

$$\frac{-0.42}{8.617 \times 10^{-5}} \times (-1.779 \times 10^{-3}) = 8.67$$

$$\frac{I_{\max}}{I_{\min}} = 4.99 \times e^{8.67} = 4.99 \times 5829 \approx \boxed{2.9 \times 10^4}$$

Problem 9

1. **Zener (BTBT) Breakdown.** In heavily-doped junctions, the depletion width is narrow. Sufficient reverse bias aligns the p-side valence band with the n-side conduction band; electrons tunnel directly through the potential barrier. Dominant at $|V_{BR}| < 5$ V. Negative temperature coefficient.
2. **Avalanche Breakdown.** In lightly-doped junctions with wide depletion regions, carriers accelerated by the electric field gain kinetic energy $\geq E_G$ and ionize lattice atoms via impact ionization, triggering a carrier multiplication cascade. Dominant at $|V_{BR}| > 7$ V. Positive temperature coefficient.

Problem 10

(a) N_A and N_D

Band offset under forward bias = $q(V_{bi} - V_A) = 0.5$ eV:

$$V_{bi} = 0.5 + 0.3 = 0.8 \text{ V}$$

For $N_A = N_D = N$:

$$V_{bi} = \frac{2kT}{q} \ln\left(\frac{N}{n_i}\right) \Rightarrow 0.8 = 0.052 \ln\left(\frac{N}{10^{10}}\right) \Rightarrow \frac{N}{10^{10}} = e^{15.38}$$

$$\boxed{N_A = N_D \approx 4.8 \times 10^{16} \text{ cm}^{-3}}$$

(b) $E_{i,p} - F_p$ and $F_n - E_{i,n}$

$$E_{i,p} - F_p = kT \ln\left(\frac{N_A}{n_i}\right) = 0.026 \times 15.38 = 0.400 \text{ eV}$$

By symmetry ($N_A = N_D$):

$$\boxed{E_{i,p} - F_p = F_n - E_{i,n} = 0.400 \text{ eV}}$$

(c) Quasi-Fermi Levels

Under the ideal diode assumptions:

- F_n is flat in the n quasi-neutral region and across the depletion region.
- F_p is flat in the p quasi-neutral region and across the depletion region.
- $F_n - F_p = qV_A = 0.3$ eV throughout the junction.
- Depletion region edges marked with dashed vertical lines; F_n lies 0.40 eV above $E_{i,n}$ in n-bulk; F_p lies 0.40 eV below $E_{i,p}$ in p-bulk.

(d) Maximum Electric Field

$$\begin{aligned} |\mathcal{E}_{max}| &= \frac{2(V_{bi} - V_A)}{W}, & W &= 2\sqrt{\frac{\varepsilon_{Si}(V_{bi} - V_A)}{qN}} \\ \varepsilon_{Si} &= 11.8 \times 8.854 \times 10^{-14} = 1.045 \times 10^{-12} \text{ F/cm} \\ \frac{\varepsilon_{Si}(V_{bi} - V_A)}{qN} &= \frac{1.045 \times 10^{-12} \times 0.5}{1.602 \times 10^{-19} \times 4.82 \times 10^{16}} = 6.77 \times 10^{-11} \text{ cm}^2 \\ W &= 2\sqrt{6.77 \times 10^{-11}} = 1.645 \times 10^{-5} \text{ cm} \\ |\mathcal{E}_{max}| &= \frac{2 \times 0.5}{1.645 \times 10^{-5}} = \boxed{6.1 \times 10^4 \text{ V/cm}} \end{aligned}$$

Problem 11

From CV curve: $C_{ox} = 20$ pF, $C_{min} = 5$ pF, $V_{FB} \approx 2$ V, area = 10^{-3} cm².

(a) Q_{ss}/q

$$V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}/A} \Rightarrow Q_{ss} = (\phi_{ms} - V_{FB}) \frac{C_{ox}}{A} = (0.2 - 2.0) \times \frac{20 \times 10^{-12}}{10^{-3}} = -3.6 \times 10^{-8} \text{ C/cm}^2$$

$$\boxed{\frac{Q_{ss}}{q} = \frac{-3.6 \times 10^{-8}}{1.602 \times 10^{-19}} = -2.25 \times 10^{11} \text{ cm}^{-2}}$$

(b) $C_{d,t}$

$$\frac{1}{C_{d,t}} = \frac{1}{C_{min}} - \frac{1}{C_{ox}} = \frac{1}{5} - \frac{1}{20} = \frac{3}{20} \Rightarrow \boxed{C_{d,t} = \frac{20}{3} \approx 6.67 \text{ pF}}$$

(c) **High or Low Frequency?**

High-frequency. In strong inversion, minority carriers cannot respond to the AC signal, so the inversion charge does not contribute to capacitance. The curve saturates at C_{min} rather than rising back to C_{ox} .

(d) **MOSFET CV with Source/Drain Grounded, $V_G > V_t$**

The source/drain contacts supply inversion charge at any frequency. For $V_G > V_t$ the capacitance rises back to C_{ox} , unlike the MOS cap case.

(e) **Inverse Subthreshold Slope**

$$S = \frac{kT}{q} \ln(10) \left(1 + \frac{C_{d,t}}{C_{ox}} \right) = 60 \left(1 + \frac{6.67}{20} \right) = 60 \times 1.333 = \boxed{80 \text{ mV/dec}}$$

(f) **ΔV_G for 10^6 reduction in I_D**

$$\Delta V_G = S \times 6 \text{ dec} = 80 \times 6 = \boxed{480 \text{ mV}}$$

Problem 12

Short-channel device. I_D continues to increase with V_{DS} in the saturation region (channel-length modulation): as V_{DS} increases, the drain-side depletion region shortens L_{eff} , raising $I_D \propto 1/L_{eff}$. A long-channel device would show flat I_D in saturation.

Problem 13

As t_{ox} decreases, $C_{ox} = \varepsilon_{ox}/t_{ox}$ increases.

(a) Flat-Band Voltage – Decrease

$$V_{FB} = \phi_{ms} - \frac{Q_{ss}}{C_{ox}}$$

$Q_{ss} < 0$ so $-Q_{ss}/C_{ox} > 0$. As $C_{ox} \uparrow$, this positive term shrinks $\Rightarrow V_{FB}$ decreases.

(b) Threshold Voltage – Decrease

$$V_t = V_{FB} + 2\phi_F + \frac{|Q_{dep,max}|}{C_{ox}}$$

Both V_{FB} and $|Q_{dep,max}|/C_{ox}$ decrease as C_{ox} increases $\Rightarrow V_t$ decreases.

(c) Inverse Subthreshold Slope – Decrease

$$S = \frac{kT}{q} \ln(10) \left(1 + \frac{C_{d,t}}{C_{ox}} \right)$$

As $C_{ox} \uparrow$, $C_{d,t}/C_{ox} \downarrow \Rightarrow S$ decreases toward the ideal limit of 60 mV/dec.

Problem 14

DIBL. In short-channel devices, the drain depletion region penetrates toward the source, partially controlling the source-channel barrier via V_{DS} :

1. V_t decreases with increasing V_{DS} ($\Delta V_t = -\eta \Delta V_{DS}$).
2. Off-state leakage I_{off} increases, degrading the on/off ratio.

Problem 15

GIDL. At the gate-drain overlap in the off-state ($V_G \approx 0$, $V_D = V_{DD}$), the large electric field causes band-to-band tunneling: electrons tunnel from valence to conduction band in the drain, generating electron-hole pairs. The resulting current (drain to substrate) increases I_{off} .

Problem 16

$T = 373$ K. Boltzmann limit (ideal FET):

$$S_{min} = \frac{kT}{q} \ln(10) = \frac{(1.381 \times 10^{-23})(373)}{1.602 \times 10^{-19}} \times 2.303 = 32.15 \text{ mV} \times 2.303 = 74.0 \text{ mV/dec}$$

$$|\Delta V_G|_{min} = 74.0 \times 5 \text{ dec} = \boxed{370 \text{ mV}}$$

Problem 17

Direct source-to-drain quantum mechanical tunneling. At $L \leq 5$ nm, the off-state potential barrier width is comparable to the electron de Broglie wavelength ($\sim 1-2$ nm in Si). The WKB tunneling probability $T \propto e^{-2\kappa L}$ ($\kappa = \sqrt{2m^*(E_B - E)}/\hbar$) is no longer negligible, producing a leakage current that cannot be suppressed by the gate. In the band diagram: even with E_c^{ch} raised above E_F^{source} by the gate, the thin barrier allows electrons to tunnel directly from source to drain.

Problem 18

$V_{GS} - V_t = 4$ V, $V_{DS} = 2$ V < 4 V $\Rightarrow$ linear regime.

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}} = \mu_n C_{ox} \frac{W}{L} V_{DS} \Rightarrow \mu_n = \frac{g_m L}{C_{ox} W V_{DS}}$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.854 \times 10^{-14}}{300 \times 10^{-7}} = 1.151 \times 10^{-8} \text{ F/cm}^2$$

$$\mu_n = \frac{(1.15 \times 10^{-6})(4 \times 10^{-4})}{(1.151 \times 10^{-8})(2 \times 10^{-4})(2)} = \frac{4.6 \times 10^{-10}}{4.604 \times 10^{-12}} = \boxed{100 \text{ cm}^2/\text{V}\cdot\text{s}}$$

Problem 19

p-type Si MOS capacitor. Band diagram labels:

Figure	State	Key feature
(c)	Accumulation	Bands bent up at surface; $E_v \rightarrow E_F$; holes pile up
(e)	Flat-band	All bands flat; no bending
(a)	Depletion	Bands slightly bent down; $E_i > E_F$ everywhere
(d)	Weak inversion	Bands bent further; $E_i \approx E_F$ near surface
(b)	Threshold	$E_i = E_F$ at surface; $\phi_s = 2\phi_F$
(f)	Strong inversion	$E_i < E_F$ at surface; dense electron inversion layer

CV curve placement (left to right, most negative to most positive V_G):

- (c) – flat plateau at C_{ox} (accumulation)
- (e) – onset of drop from C_{ox} at V_{FB}
- (a) – mid-slope between V_{FB} and V_t
- (d) – lower slope approaching C_{min}
- (b) – $C = C_{min}$ at V_t
- (f) – flat plateau at C_{min} (strong inversion, high frequency)